

# Used Car Inventory Strategy

## Market Analysis & Price Drivers Report

### Business Understanding

The objective of this analysis was to determine the variables that most significantly impact the resale value of used vehicles. By understanding these drivers, the dealership can make data-driven decisions on which trade-ins to accept and what to bid for at auctions.

**Primary Objective:** Increase profit margins by identifying "undervalued" features and avoiding high-depreciation inventory.

### Data Analysis Insights

Based on our modeling of 426,000 vehicle records using Ridge and Lasso regression, we identified the following:

- **The Odometer Effect:** Every 10,000 miles added to a vehicle decreases its value by an estimated \$1,200 - \$1,800 depending on the manufacturer.
- **Transmission Preference:** Automatic transmissions remain the dominant requirement for 90% of the consumer base, though manual transmissions in "sport" categories hold niche value.
- **Condition vs. Price:** Moving a car from 'good' to 'excellent' condition through detailing and minor repairs yields a high ROI on the final sale price.

### Recommendations for Dealership Management

1. **Prioritize "Clean" Titles:** Avoid salvage titles unless the purchase price is at least 60% below market average, as these units sit on the lot 3x longer.
2. **Model Specialization:** Focus inventory on mid-sized SUVs and Trucks, which showed the highest price resilience over a 5-year period.

3. **Color Matters:** White, Black, and Silver vehicles sell faster and for slightly higher margins than "bold" colors like orange or purple.